4th Meeting 11/11

I already trained my model → WorkerNET → unified model (Detection, Pose, Reading Barcode)

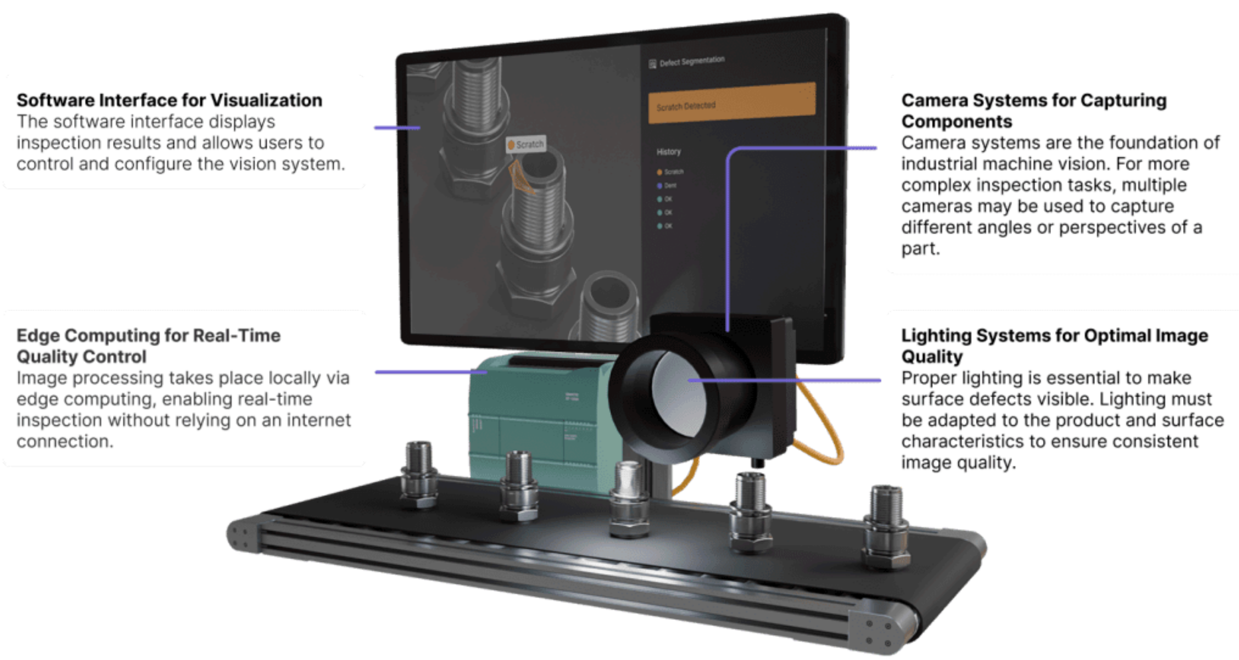
- I try to compare with ordinary Yolo v8 (Detection only) with 49927 training data and 7614 total valid samples [Bottle, Barcode, People] Dataset

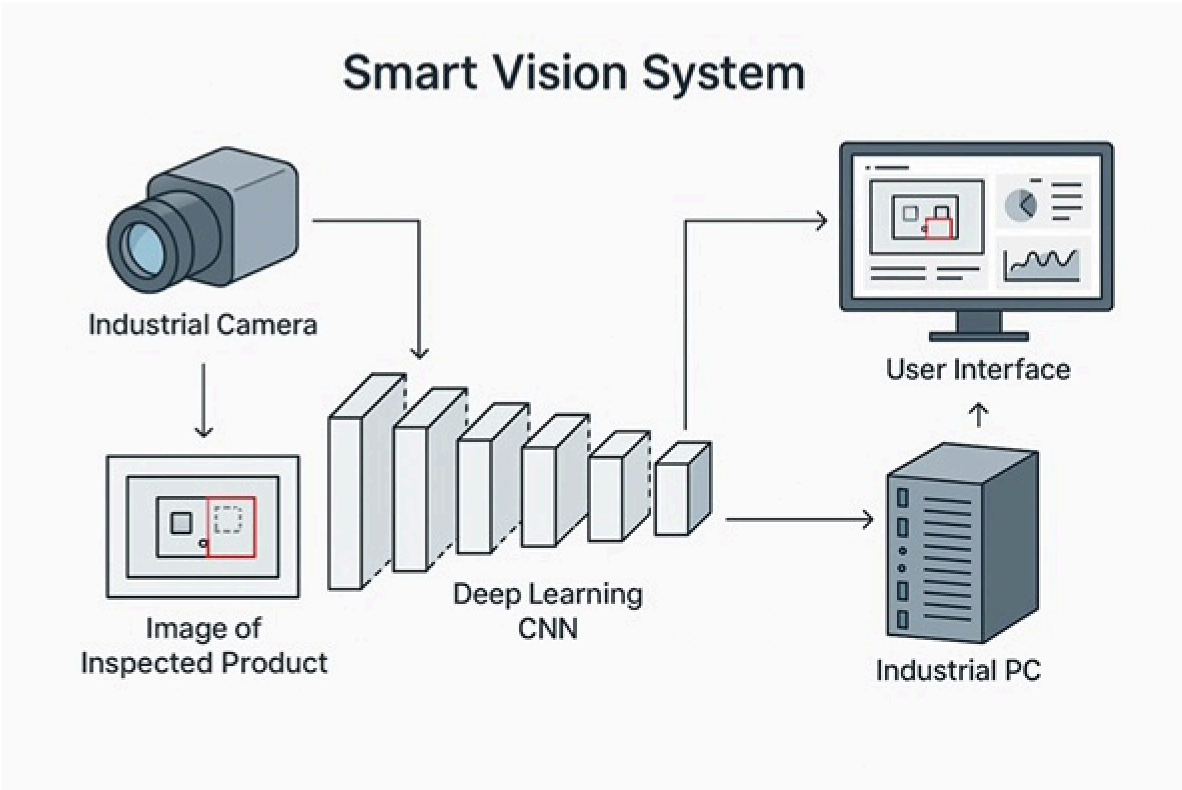
```
    Barcode/data1: 1673 samples
    Barcode/data2: 1126 samples
    Barcode/data3: 1674 samples
    Barcode/data4: 1673 samples
    Barcode/data5: 28696 samples
    Bottle/data1: 275 samples
    Bottle/data2: 257 samples
    Bottle/data3: 1972 samples
    Bottle/data4: 1438 samples
    Bottle/data5: 1972 samples
    Bottle/data6: 72 samples
    Bottle/data7: 257 samples
    Person/data3: 1400 samples
    Person/data4: 3721 samples
    Person/data2: 3721 samples

  Total train samples: 49927
```

```
--- Epoch 1/100 ---
Epoch 1 Training:   8%|   | 239/3120 [00:14<02:43, 17.64it/s, loss=36.3, det=43.5, barcode=2.29, pose=3.64e-6]
Corrupt JPEG data: premature end of data segment
Epoch 1 Training:  16%|   | 499/3120 [00:29<02:28, 17.67it/s, loss=28.3, det=43.9, barcode=2.3, pose=1.19e-6]
Corrupt JPEG data: 32 extraneous bytes before marker 0xd9
Epoch 1 Training:  45%|   | 1419/3120 [01:23<01:39, 17.09it/s, loss=12, det=44.6, barcode=2.3, pose=1.51e-7]
Corrupt JPEG data: premature end of data segment
Epoch 1 Training:  50%|   | 1551/3120 [01:30<01:31, 17.13it/s, loss=8.23, det=33, barcode=2.3, pose=1.19e-8]
Corrupt JPEG data: 300 extraneous bytes before marker 0xd9
Epoch 1 Training:  85%|   | 2663/3120 [02:35<00:27, 16.88it/s, loss=5.48, det=46, barcode=2.31, pose=8.37e-10]
Corrupt JPEG data: bad Huffman code
Epoch 1 Training: 100%|   | 3120/3120 [03:01<00:00, 17.17it/s, loss=5.06, det=54.8, barcode=2.31, pose=1.24e-9]
Epoch 1 avg_train_loss: 14.653672
```

```
--- Generating Comparison Report ---
Time Comparison:
  Traditional: 19950.12s
  WorkerNet:   0.00s
  Time Saved:  19950.12s (100.00%)

Performance Comparison (mAP50):
  Traditional - barcode: 0.9777
  Traditional - bottle: 0.8879
  Traditional - person: 0.6252
  Traditional - pose: 0.0000
  WorkerNet (Unified): 0.0000
```

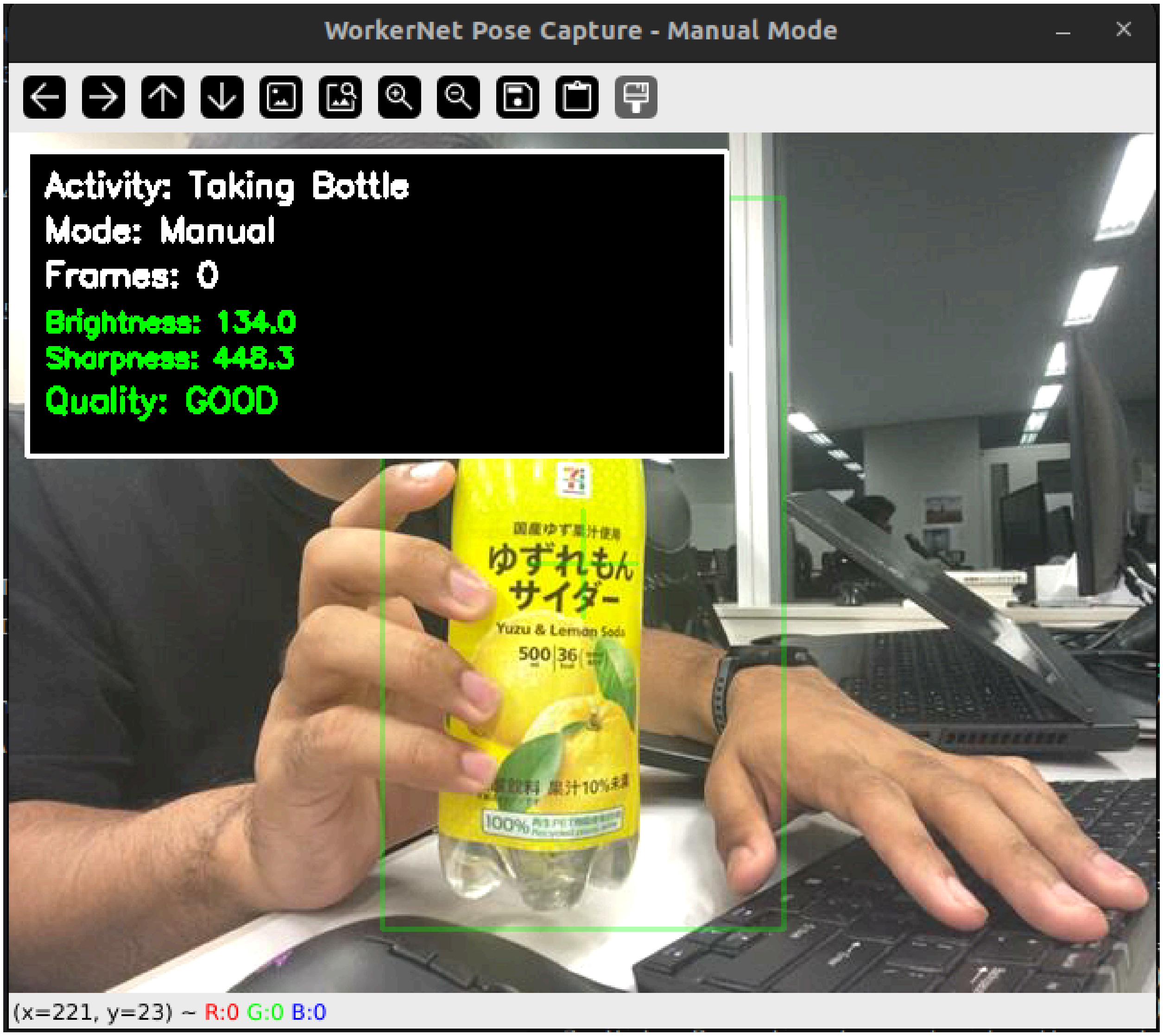
I need make sure all of the pose identification is correctly recognized in each activity → So I will make an UI to verify each step of the process to ensure it's done based on procedure





Other than that, I will feed the model with my own dataset that I annotate myself to familiar with real condition of the lab


ゆずれもん
サイダー



Keypoint 0: nose
Keypoint 1: left_eye
Keypoint 2: right_eye
Keypoint 3: left_ear
Keypoint 4: right_ear
Keypoint 5: left_shoulder
Keypoint 6: right_shoulder
Keypoint 7: left_elbow
Keypoint 8: right_elbow
Keypoint 9: left_wrist

Keypoint 10: right_wrist
Keypoint 11: left_hip
Keypoint 12: right_hip
~~Keypoint 13: left_knee~~
~~Keypoint 14: right_knee~~
~~Keypoint 15: left_ankle~~
~~Keypoint 16: right_ankle~~

Check the cap is already closed or no

For this semester → focus on finish the whole prototype process to be able to join the conference → i can use this idea to my phd dissertation to talk about more complex thing (implementation in the real company(?))

Focus on this semester → focus on the workflow and prove this method is possible

Contract of the job

They will get 10 yen every bottle being checked

in the end of the shift, i will sum all of the checked bottle (if example 100 bottles, then i will transfer 1000 yen directly to the wallet blockchain of the workers automatically)

Check the bottle → recognize its barcode (to get bottle ID) → verify if the cap is already closed (it is correct one) → counting the numbers of the checked and correct bottle → calculate all of the bottle that being checked → store it to blockchain → transfer the compensation/payroll to the employee/workers based on the job done

1. ****Taking Bottle**** - Worker reaching and grasping bottle

2. ****Scanning Barcode**** - Worker positioning bottle for barcode scan
3. ****Checking Bottle**** - Worker inspecting bottle quality/defects
4. ****Checking Cap**** - Worker verifying cap closure
5. ****Storing Bottle**** - Worker placing bottle in storage/rack
6. ****Bottle Rotation**** - Worker rotating bottle for inspection

Expected Timeline

Just to make sure the whole process is done correctly

Small Dataset (1,500 frames):

- Recording: 2-3 hours
- Frame extraction: 30 minutes
- Annotation: 15-20 hours
- Training: 2-3 hours

Total: ~25 hours

Medium Dataset (5,000 frames):

- Recording: 5-7 hours
- Frame extraction: 1 hour
- Annotation: 50-70 hours
- Training: 4-6 hours

Total: ~70 hours

Large Dataset (10,000 frames):

- Recording: 10-15 hours
- Frame extraction: 2 hours

- Annotation: 100-150 hours
- Training: 8-12 hours

Total: ~150 hours

Short Term Planning (2)

Name	Date	Tags
Untitled		
Write paper and submission	@December 31, 2025 → January 31, 2026	
Address feedback, revise code and documentation		
Prepare figures, diagrams, and tables for paper		
POPW Implementation COMPLETE	@December 15, 2025 → December 31, 2025	
Address feedback, revise code and documentation		
Prepare figures, diagrams, and tables for paper		
Data and result collection for paper	@December 8, 2025 → December 14, 2025	
Address feedback, revise code and documentation		
Prepare figures, diagrams, and tables for paper		
Optimization	@December 2, 2025 → December 8, 2025	
Real world lab testing	@November 25, 2025 → December 1, 2025	
Debugging	@November 19, 2025 → December 1, 2025	
Untitled		
POPW protocol implementation	@November 11, 2025 → November 30, 2025	
Implement the spatiotemporal fingerprint	@November 11, 2025 → November 30, 2025	
Data Collection	@November 11, 2025 → November 30, 2025	

Name	Date	Tags
Finalize architecture and prepare result	@October 31, 2025 → November 30, 2025	
Complete workernet training & benchmarking	@October 14, 2025 → November 30, 2025	